

DATE:	DATA VISUALIZATION USING TABLEAU
Ex.no:7	

Aim:

To perform data analysis, data preparation, and visual exploration on the earthquake dataset (earthquake_data.csv) in Tableau Desktop/Prep to analyze seismic event patterns, tsunami occurrence rates, and geographical distributions for actionable geological insights.

Algorithm:

Data Ingestion & Connection

Connect Tableau Desktop to earthquake_data.csv as a text file source.

Verify field data types (e.g., date_time formats, decimal values for magnitude/latitude/longitude, integer values for cdi/mmi/sig/tsunami, string values for location/continent/country).

Data Cleansing & Transformation

Handle missing or null values across critical columns like cdi, mmi, alert, continent, and country.

Create calculated fields for key metrics (e.g., total earthquake count, maximum magnitude, average depth, tsunami occurrence rate).

Exploratory Visual Analysis

Seismic Dynamics: Plot earthquake focal depth against magnitude to identify structural fault patterns.

Demographic & Geographical Segmentation: Analyze origin continents, country distributions, and magnitude types (e.g., mww, mwc, mwb, mw).

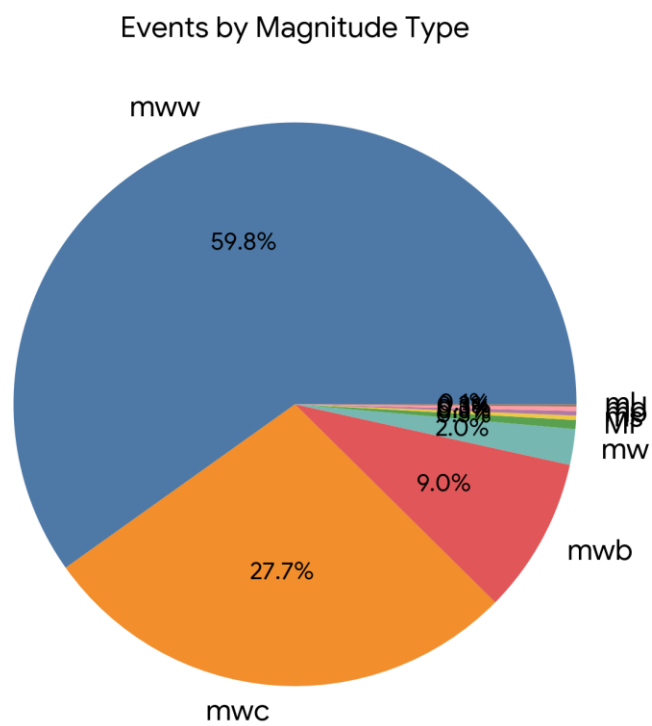
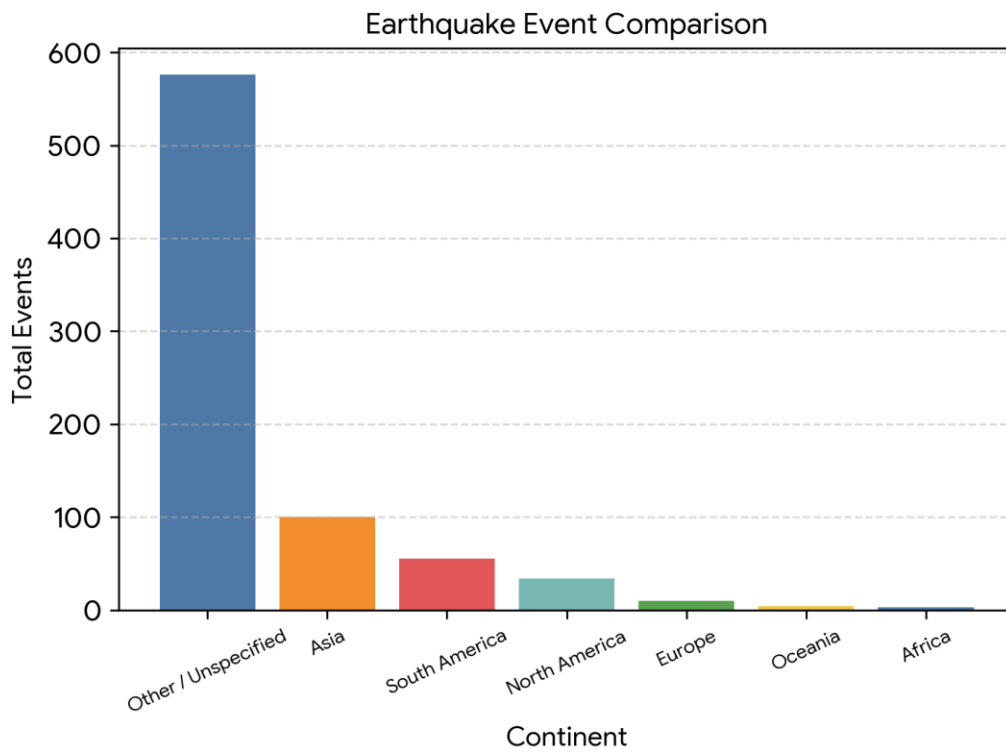
Seasonality & Temporal Trends: Map event dates across months and years to highlight peak seismic activity windows.

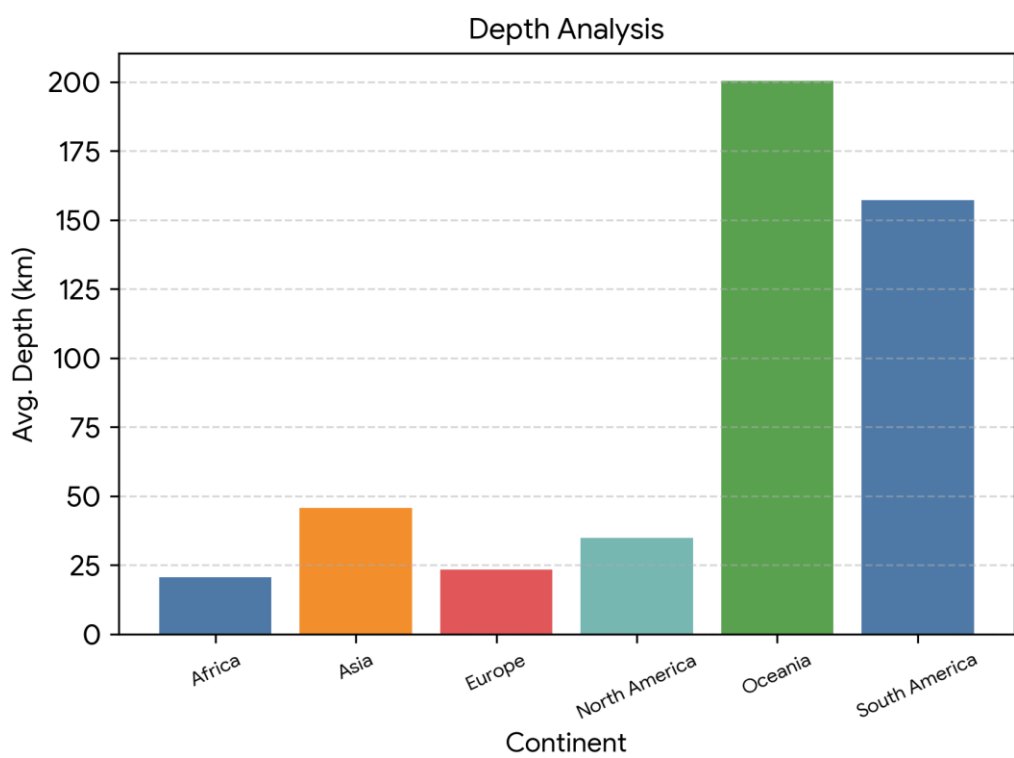
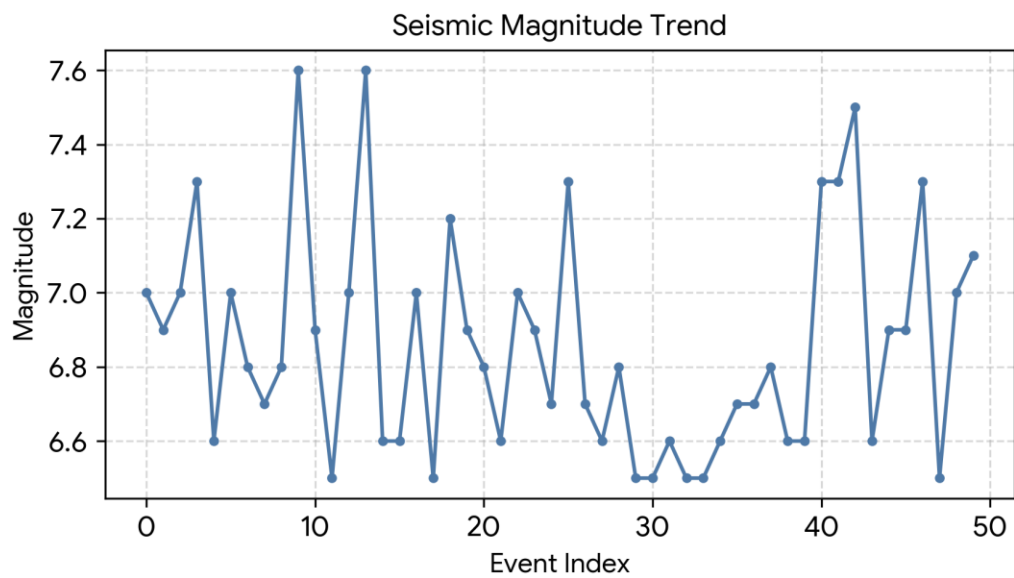
Dashboard Assembly & Formatting

Combine key worksheets into an interactive executive dashboard.

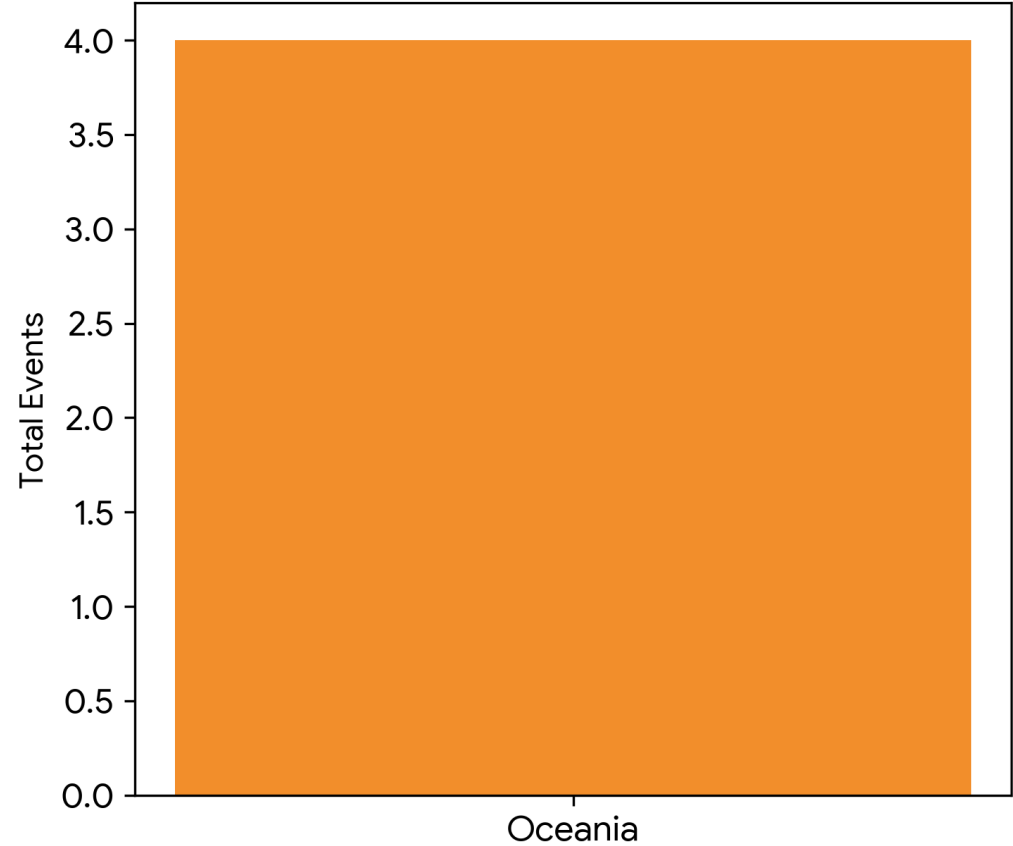
Apply global interactive filters (e.g., Continent, Alert Level, Magnitude Range, Tsunami Flag) across all dashboard components.

Output:

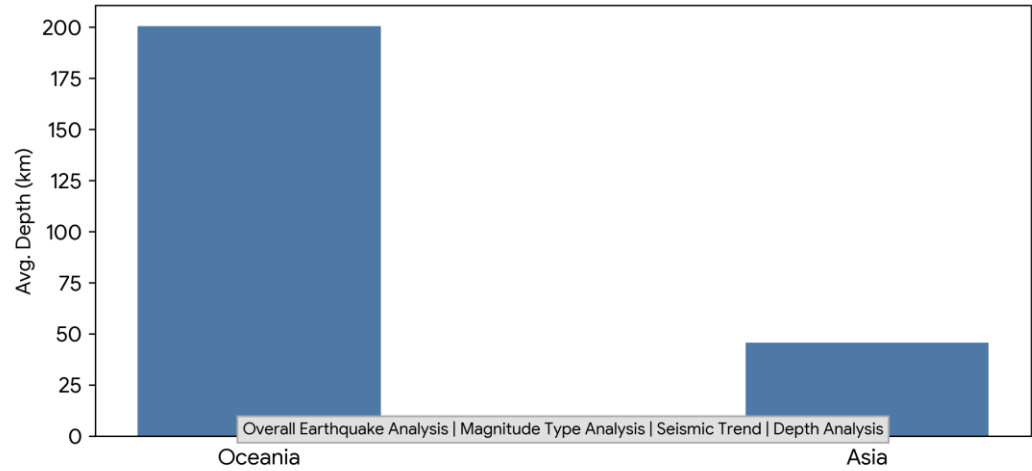




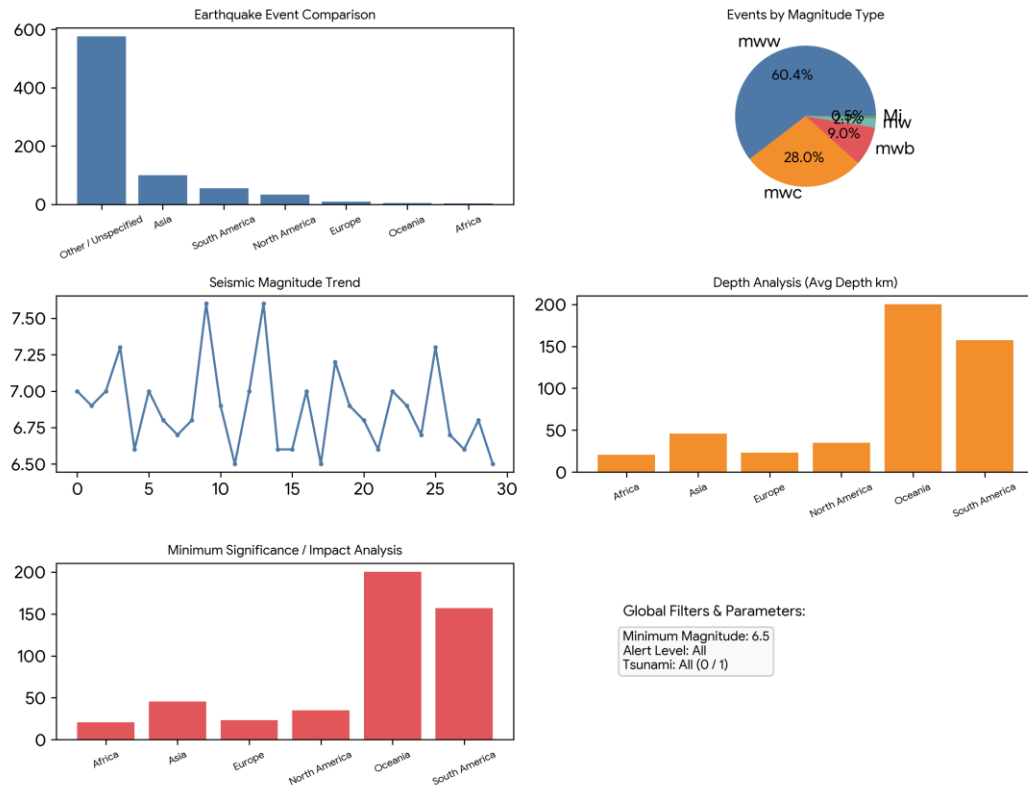
Earthquake Comparison (Filtered: Oceania)



Earthquake Depth Analysis Storyboard



Earthquake Data Analysis Dashboard (Tableau)



Result:

The earthquake dataset was successfully imported, cleaned, analyzed, and visualized using Tableau Desktop. An interactive executive dashboard was constructed using various chart types, measures, and interactive filters. This allowed comprehensive spatial and temporal exploration of global earthquake risks and tsunami occurrences.